



Arthur Fyon

ID: 595-0556300-23 | **Date of birth:** 12/03/1999 | **Place of birth:** Verviers, Belgium |

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Address: Allée de la Découverte 10, 4000, Liège, Belgium (Work)

● EDUCATION AND TRAINING

15/09/2021 – 13/02/2025 Liège, Belgium

DOCTORATE IN ENGINEERING SCIENCES AND TECHNOLOGY (ELECTRICAL, ELECTRONIC AND COMPUTER ENGINEERING) ULiège

Website <https://www.uliege.be/> | **Field of study** Engineering, manufacturing and construction | **Type of credits** ECTS |

Number of credits 60 |

Thesis Uncovering the molecular and cellular mechanisms of reliable neuromodulation in highly heterogeneous neurons

15/09/2019 – 30/06/2021 Liège, Belgium

MASTER OF SCIENCE (MSC) IN BIOMEDICAL ENGINEERING ULiège

Website <https://www.uliege.be/> | **Field of study** Engineering, manufacturing and construction |

Final grade Summa cum laude with congratulations of the jury | **Type of credits** ECTS | **Number of credits** 120 |

Thesis A wearable miniaturized wireless multisensor of physiological signals for continuous monitoring

15/09/2016 – 30/06/2019 Liège, Belgium

BACHELOR OF SCIENCE (BSC) IN ENGINEERING ULiège

Website <https://www.uliege.be/> | **Field of study** Engineering, manufacturing and construction | **Final grade** Summa cum laude |

Type of credits ECTS | **Number of credits** 180

● WORK EXPERIENCE

POSTDOCTORAL RESEARCHER – FONDS DE LA RECHERCHE SCIENTIFIQUE - FNRS – 14/02/2025 – 30/09/2026 – LIÈGE, BELGIUM

Department: Department of Electrical Engineering and Computer Science

Postdoctoral researcher in the Neuroengineering Lab of the ULiège under grant ASP-REN40024838. My research topics are computational neuroscience, bio-inspired machine learning and neuromorphic electronics.

UNIVERSITY TEACHING ASSISTANT – ULIÈGE – 15/09/2021 – 14/09/2022 – LIÈGE, BELGIUM

Department: Department of Electrical Engineering and Computer Science

Teaching assistant in the following courses:

- ELEN0040 - Digital Electronics: supervision of laboratory sessions and projects
- SYST0003 - Linear Control Systems: supervision of exercise sessions and projects
- SYST0017 - Neurodynamics: supervision of exercise sessions and projects

Student instructor in the following courses:

- MATH0062 - Elements of Probability Calculus: drafting and editing of projects
- ELEC0052 - Electric Measurements: Foundations and Applications: supervision of laboratory sessions
- CHIM0286 - Rudiments of Thermodynamics: supervision of exercise sessions

● LANGUAGE SKILLS

Mother tongue(s): **FRENCH**

Other language(s):

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken production	Spoken interaction	
ENGLISH	C2	C2	C1	C1	C2

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

● SKILLS

Microsoft Office | Microsoft Word | Microsoft Excel | Microsoft Powerpoint | LaTeX | Outlook | Google Drive | Simulink | Inkscape | Windows

Programming languages

Python | Julia | Arduino | Git | Linux | VHDL | C | MATLAB | Microchip assembly language | Java

Simulation and design softwares

Altium Designer | Gmsh/GetDP | LTSpice | Cadence Virtuoso

● HONOURS AND AWARDS

16/10/2026

AIM Best PhD Thesis – AIM

Best Thesis Award for 2025, awarded by the Department of Electrical Engineering and Computer Science at the University of Liège. The award includes a €3,500 prize.

28/06/2022

Research Fellow Grant – Fonds de la Recherche Scientifique - FNRS

The Research Fellow mandate aims to complete a doctorate in four years. It takes the form of a two-year scholarship, which may be renewed for a maximum of two years.

27/11/2021

Best Master Thesis in Biomedical Engineering: Trasis Award – FACSA ULiège & Trasis

“In recognition of the academic excellence of the final thesis, as acknowledged by the joint juries of the Faculty of Applied Sciences at the University of Liège.”

28/08/2016

Pisart Merit Scholarship – Legs Fernand Pisart

● PUBLICATIONS

2023

[Reliable neuromodulation from adaptive control of ion channel expression](#)

Fyon, A., Sacré, P., Franci, A., & Drion, G. (2023). Reliable neuromodulation from adaptive control of ion channel expression. IFAC-PapersOnLine, 56(2), 458-463.

Authors: Arthur FYON, Pierre SACRE, Alessio FRANCI and Guillaume DRION | **Journal Name:** IFAC-PapersOnLine | **Volume, Issue and Pages:** Vol. 56 Issue 2, p458-463 | **Publisher:** Elsevier

2024

[Dimensionality reduction of neuronal degeneracy reveals two interfering physiological mechanisms](#)

Fyon, A., Franci, A., Sacré, P., & Drion, G. (2024). Dimensionality reduction of neuronal degeneracy reveals two interfering physiological mechanisms. PNAS nexus, 3(10), pgae415.

Authors: Arthur FYON, Alessio FRANCI, Pierre SACRE and Guillaume DRION | **Journal Name:** PNAS Nexus | **Volume, Issue and Pages:** Vol. 3 Issue 10, pgae415 | **Publisher:** National Academy of Sciences

2024

[Controlled neuromodulation naturally interacts with homeostasis to produce robust and modulable neuronal function](#)

Fyon, A., & Drion, G. (2024). Controlled neuromodulation naturally interacts with homeostasis to produce robust and modulable neuronal function. arXiv preprint, 2412.04172.

Authors: Arthur FYON and Guillaume DRION | **Journal Name:** arXiv preprint | **Volume, Issue and Pages:** 2412.04172

2025

[A fully tunable ultra-low power current-mode memory cell in standard cmos technology](#)

Fyon, A.*, Mendolia, L.*, et al. (2025). A fully tunable ultra-low power current-mode memory cell in standard cmos technology. ORBi preprint, 2268/338452.

Authors: Arthur FYON*, Loris MENDOLIA*, Jean-Michel REDOUTE, Alessio FRANCI and Guillaume DRION | **Journal Name:** ULiège institutional repository ORBi preprint | **Volume, Issue and Pages:** 2268/338452

2025

[Fast reconstruction of degenerate populations of conductance-based neuron models from spike times](#)

Brandoit, J., Ernst, D., Drion, G., & Fyon, A. (2025). Fast reconstruction of degenerate populations of conductance-based neuron models from spike times. arXiv preprint, 2509.12783.

Authors: Julien BRANDOIT, Damien ERNST, Guillaume DRION and Arthur FYON | **Journal Name:** arXiv preprint | **Volume, Issue and Pages:** 2509.12783

2025

[A fast and tiny current as common generator of slow regular pacemaking in brain and heart](#)

Fyon, A., et al. (2025). A fast and tiny current as common generator of slow regular pacemaking in brain and heart. bioRxiv preprint, 10.1101/2025.10.30.685563v2.

Authors: Arthur FYON, Oleksandra PAVLOVA, Sofian RINGLET, Alessio FRANCI, Pietro MESIRCA, Matteo MANGONI, Nick SCHAAR, Jochen ROEPER, Vincent SEUTIN*, Guillaume DRION* and Kevin JEHASSE* | **Journal Name:** bioRxiv preprint | **Volume, Issue and Pages:** 10.1101/2025.10.30.685563v2

2026

[Co-design of AI for resource-constrained environments](#)

Fyon, A., et al. (2026). Co-design of ai for resource-constrained environments. ORBi preprint, 2268/339660.

Authors: Arthur FYON, Julien BRANDOIT, Damien ERNST, Jean-Michel REDOUTE and Guillaume DRION | **Journal Name:** ULiège institutional repository ORBi preprint | **Volume, Issue and Pages:** 2268/339660

2026

[Improving the performance and learning stability of parallelizable rnns designed for ultra-low power applications](#)

Brandoit, J., Fyon, A., Ernst, D., & Drion, G. (2026). Improving the performance and learning stability of parallelizable RNNs designed for ultra-low power applications. ORBi preprint, 2268/339664.

Authors: Julien BRANDOIT, Arthur FYON, Damien ERNST and Guillaume DRION | **Journal Name:** ULiège institutional repository ORBi preprint | **Volume, Issue and Pages:** 2268/339664

2026

[Tunable current-mode bistable switching circuit](#)

Fyon, A., Mendolia, L., Franci, A., & Drion, G. (in press). Tunable Current-Mode Bistable Switching Circuit.

Authors: Arthur FYON, Loris MENDOLIA, Alessio FRANCI and Guillaume DRION | **Journal Name:** Patent in press

● CONFERENCES AND SEMINARS

08/11/2021 – 11/11/2021 Online

Apparent neuromodulation-dependent correlation levels in neuron ion channel expression result from a competition between passive and active membrane properties

Poster presentation at the Society for Neuroscience (SfN) Annual Meeting 2021 (Online)

09/07/2023 – 14/07/2023 Yokohama, Japan

Reliable neuromodulation from adaptive control of ion channel expression

Oral presentation at the 22nd International Federation of Automatic Control (IFAC) World Congress (Yokohama)

26/09/2023 – 29/09/2023 Berlin, Germany

Neuromodulation and homeostasis: complementary mechanisms in ion channel expression adaptation for robust neural function

Poster presentation at the Bernstein Conference 2023 (Berlin)

25/06/2024 – 29/06/2024 Vienna, Austria

Neuromodulation robustly tunes rhythmic patterns in a fixed connectome: application to gait control

Poster presentation at the Federation of European Neuroscience Societies (FENS) Forum 2024 (Vienna)

26/03/2024 – 28/03/2024 Blankenberge, Belgium

An adaptive controller of reliable neuromodulation on mixed feedback systems

Oral presentation at the 43rd Benelux Meeting (Blankenberge)

29/11/2024 – 29/11/2024 Brussels, Belgium

An experiment-based robust model for slow pacemaking

Oral presentation at the Autumn Meeting 2024 of the Belgian Society of Physiology and Pharmacology (Brussels)

18/03/2025 – 20/03/2025 Egmond Aan Zee, The Netherlands

Neuromodulation and homeostasis: complementary controllers for robust neural function

Oral presentation at the 44th Benelux Meeting (Egmond Aan Zee)

14/09/2025 – 17/09/2025 Sète, France

Slow dynamics from fast activating kinetics in pacemaker cells

Poster presentation at the 34th Ion channels meeting (Sète)

15/11/2025 – 19/11/2025 San Diego, U.S.A.

Slow pacemaking dynamics from fast activating kinetics in midbrain dopamine neurons

Poster presentation at the Society for Neuroscience (SfN) Annual Meeting 2025 (San Diego)

28/11/2025 – 28/11/2025 Brussels, Belgium

A tiny and fast current as common generator for regular slow pacemaking in brain and heart

Oral presentation at the Autumn Meeting 2025 of the Belgian Society of Physiology and Pharmacology, Brussels.

07/12/2025 – 07/12/2025 San Diego, U.S.A.

Fast reconstruction of degenerate populations of conductance-based neuron models from spike times using deep learning

Poster presentation at the 39th Annual Conference on Neural Information Processing Systems (NeurIPS) - Workshop : Data on the Brain & Mind (San Diego).

12/03/2026 – 15/03/2026 Lisbon, Portugal

Neuronal population inference reveals how neuromodulation reshapes conductance space

Poster presentation at the Computational and Systems Neuroscience (COSYNE) 2026, Lisbon.

24/03/2026 – 26/03/2026 Lommel, Belgium

Slow neuronal dynamics from fast activating kinetics in conductance-based models

Oral presentation at the 45th Benelux Meeting on Systems and Control, Lommel.

24/03/2026 – 26/03/2026 Lommel, Belgium

Inferring neuromodulatory targets and conductance parameters from neuronal spike times

Oral presentation at the 45th Benelux Meeting on Systems and Control, Lommel.